

A Low-Power High-Precision Time-Domain Winner-Take-All Circuit: Theory and Implementation

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Abstract—In several sensors and machine learning applications, there are often a need to select the highest element of a sequence of current or voltage. This function can be achieved by a circuitry called winner-take-all (WTA) circuit. In this paper, an improvement on previous work using the time-domain winner-take-all circuit structure is proposed. The new proposed circuit achieves 99.5% accuracy and $0.15 \mu W$ at 1 MHz. The circuit structure leverages voltage to time conversion as well as positive latch circuitry to achieve fast and accurate decision making at low voltage, which reduces overall power consumption. This work is made possible with the TSMC $0.18 \mu m$ technology. The silicon area of the proposed circuit is $26 \mu m \times 40 \mu m$.

Keywords—Winner-Take-All (WTA), low power, subthreshold, analog computing

I. INTRODUCTION

Winner-Take-All (WTA) circuits that selects the highest current or voltage from a sequence has seen a rise in popularity in recent years in research. This trend is partially due to the rise of machine learning applications as well as the need for high accuracy and high speed image sensors due to its ability to conduct fast neuron spikes like behaviour in a low power and high accuracy way. In a recent research in the field of image sensing, the author used a 10 MHz WTA with a decision time of $1.6 \mu s$ to achieve fast localisation during optical triangulation [1]. The utilisation of WTA enabled the ability of image sensing circuits to quickly identify active pixels in an image which enables applications such as fast 3D imaging. Another interesting application of WTA has been in the field of machine learning. Here, WTAs are used in multiple applications such as a Hamming distance discriminator circuits presented in [2] where the WTA circuit is used to select elements with smallest hamming distances. Further applications of the WTA circuits also concern the construction of full-wave rectifiers and analog switches. In these applications, two input WTAs are used to select the different mode of operation. For rectifiers, comparisons are made between ground and input signals. Whereas for analog switches, comparisons are made between digital clocks and analog signals [3].

Among all three applications, four significant factors must be considered: **power consumption, precision, number of inputs, and speed**. In machine learning applications, low power applications have been rising in interest, however, accuracy

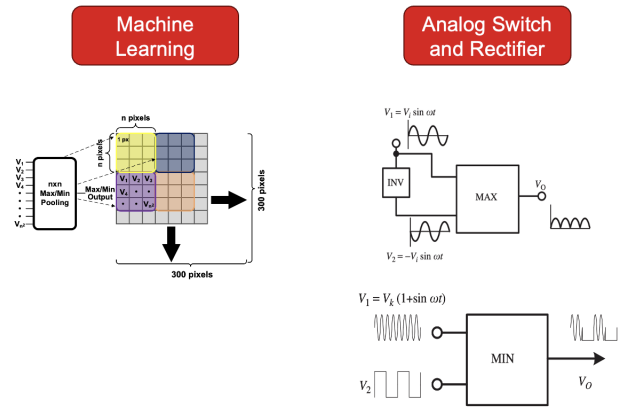


Fig. 1. Common Applications of WTA

remains an important aspect. On the other hand, for image sensors and other applications, WTA is required to be fast and high precision. With these key factors in mind, we present in this paper a $0.18 \mu m$ CMOS four input low power mixed-signal WTA that achieves 99.7% accuracy and $0.15 \mu W$ of power consumption.

This paper is organized as follows. In Section II some notions of WTA will be discussed in order to give readers a background on WTA implementations and realizations. In Section III, the figure of merits and key design aims of WTA will be discussed. The definition and calculation of these design aims will be mentioned as well. In Section IV, recently published WTA structures will be discussed and examined. In Section V, the WTA structure proposed in this work will be discussed in detail. In Section VI, we will demonstrate the simulated result as well as a discussion on power, speed, and accuracy. A comparison with state-of-the-art research will be made in Section VIII. Lastly, we will conclude the findings in Section X. During this project, part of our exploration is related to analog WTA, thus, it will be included as well in section XI.

II. NOTIONS ON WTA CIRCUITS

According to different applications, WTA circuits can appear in the form of pure analog or mixed-signals. Analog WTA

circuits are made in a way such that the output voltage follows the input voltage., i.e. $f(x_1, x_2, x_3, x_4) = x_2$ when x_2 is the winner. An example implementation can be refereed in figure 2. However, certain digital/mixed-signal implementations selects the highest voltage/current without providing information on their exact values, i.e. $f(x_1, x_2, x_3, x_4) = [0, 1, 0, 0]$ in the case where x_2 is the largest value. Implementation can be referred to in [4]. These implementations are useful in different cases. We can imagine that digital WTA can be used to identify the address of the pixel with the highest value, which allows digital processing to quickly identify the region of interest. On the other hand, applications like rectifiers or analog switches require analog WTA that can provide analog output signals. Applications like max-pooling function in machine learning can also greatly benefit from analog WTA [5].

III. FIGURE OF MERITS IN WTA CIRCUITS

As mentioned in the introduction, designer of a WTA should have the following aims in mind: precision, speed, number of inputs, and power consumption. In this section, we will discuss each aspects and their definition.

A. Precision

Out of these three aims, accuracy is a metric that might be novel for people who first got into this field. The definition of accuracy is discussed in [4] and [6]. Essentially, accuracy (γ) is defined as such:

$$\gamma := \frac{\Delta}{[V_{dynamic}, I_{dynamic}]}$$

where Δ is the resolution and the $[V_{dynamic}, I_{dynamic}]$ is the input dynamic range. Resolution Δ is defined as the minimum distance between the highest input to the second highest input.

$$\Delta := \min[V_{1st} - V_{2nd}], \min[I_{1st} - I_{2nd}]$$

The input dynamic range is the range where at least one of the inputs must be to ensure proper functioning. For example, in figure 2, the voltage dynamic range will be $[V_{I_{sat}}, V_{DD}]$.

$$V_{dynamic} := V_{in,max} - V_{in,min}$$

B. Speed

Definition of speed can be realised in different ways. In this paper, we define the speed of the circuit as the clock to output delay ($CLK - Q$). This measure is only possible in digital/mixed-signal circuits. For analog WTA, one could define it as the 10% to 90% rise/fall time of a step response.

C. Number of Inputs

Number of inputs in WTA are highly dependent on architecture. Binary-Tree type architecture can be formed to provide a larger amount of acceptable inputs, however, this decreases speed and increases input consumption. Thus, it is important to take into consideration of the amount of inputs to normalize

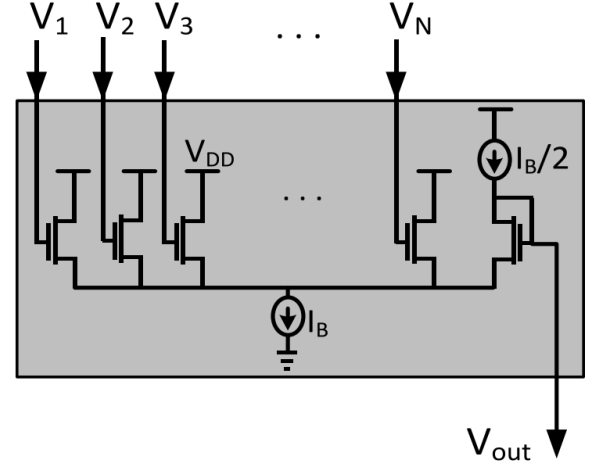


Fig. 2. A simple voltage-mode WTA circuit

metrics like power consumption for comparison between different works.

D. Power Consumption

Power consumption is the conventional calculation of $P = IV$. However, since WTA is flexible in its construction and can be expanded using a cascade binary-tree like structure, we often normalises the power to reflect the actual power consumption for each input.

$$P_{input} = \frac{I_{total} \cdot V_{DD}}{N}$$

E. Frequency

Frequency is often used as a proxy for speed. However, sometimes circuits are not run/tested at their maximal speed, thus making this parameter not as reflective as their actual operational limit. Thus, in this work, we provide both the performance of frequency as well as delay to give a holistic view of the circuit. Here, the frequency (f) is defined as the frequency of the clock used in measurements and simulation. In the case of analog WTA, where no clock is present, will simply not have frequency as part of its figure of merit.

F. Figure of Merits

From previous research, the figure of merit used is defined as such in [7].

$$FoM := \frac{Power}{f \cdot N}$$

Considering the importance of speed and precision, this paper also proposes an alternative FoM that takes into account of the two factors.

$$FoM_{new} := \frac{Power}{f \cdot N} \cdot t_{delay} \cdot \gamma$$

IV. RELEVANT WORKS

The implementation of WTA circuits originated from leveraging the non-linearity of an analog differential pair to select the higher voltage input like displayed in figure 2. Here, the source of each input transistors are connected to a common low impedance node. When an input voltage is significantly higher than another, the common-node voltage increases, thus reducing the V_{GS} voltage of the other transistors and push them into linear region. Using the low conductivity in linear region, the winning transistor (the one with winning input) takes all the bias current, which is $I_B/2$ and the output voltage is recreated using the diode connected transistor that has equal sizing and equal current.

This architecture is further improved and generalized in the [8], which is one of the original paper of analog WTA. In this paper, a feedback loop is implemented to strengthen the current winning operation. However, the concept of winning current and recreating their value remains similar to the simple WTA displayed in figure 2. Other recent analog implementations continue to build on the work in [8] by extending its capability to operate in low voltage region [9] as well as improving the speed and accuracy through reducing the impedance of the common node [10].

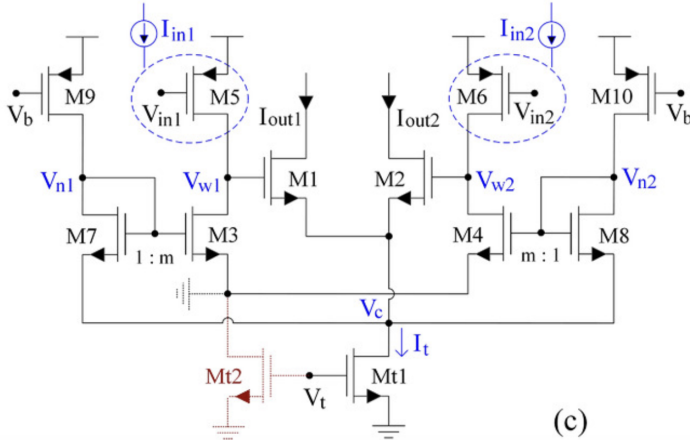


Fig. 3. Schematic of a low voltage analog WTA block presented in [9]

Digital WTA has also been proposed to achieve similar operation. Recent works in the digital WTA domains can be found in [4], [11]. In both works, the use of delay block are the key to the functioning of such circuit. In figure 5, input voltages are converted to delays that will then be latched on by a three input cross-coupled NAND gates [4]. Similar logic is used in figure 6 where the image pixels intensity is converted to delay blocks which is then passed to a flip-flop [11]. Digital WTAs have seen increase in popularity because of its ability to remain low power during its operation. Specifically, in [4], the power consumption per input reached 250nW where as the analog WTAs remains in the μW region.

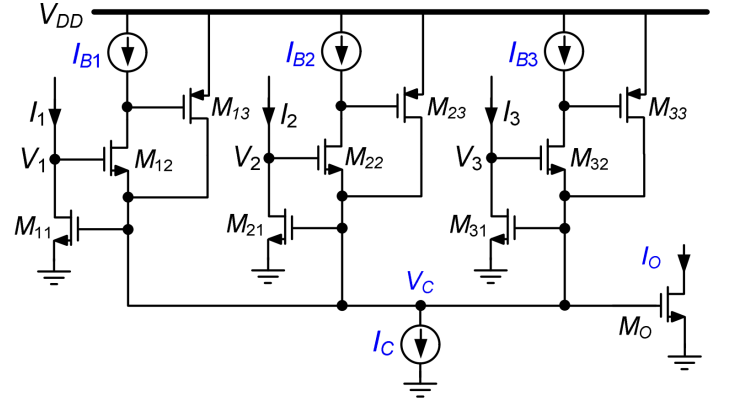


Fig. 4. Schematic of an improved Lazzazo voltage analog WTA block presented in [10]

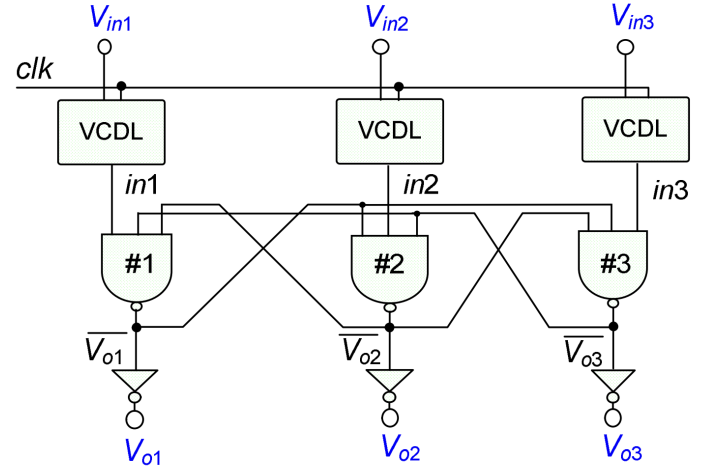


Fig. 5. Schematic of a time-domain digital WTA proposed in [4]

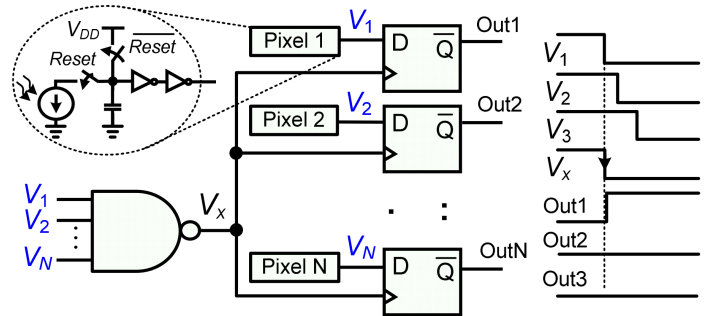


Fig. 6. Schematic of a time-domain digital WTA proposed in [11]

V. PROPOSED CIRCUIT

The proposed circuit takes the form of a voltage-mode digital/mixed-signal WTA, i.e. $f(x_1, x_2, x_3, x_4) = [0, 1, 0, 0]$ when x_2 is the highest voltage. The circuit is composed of three main blocks: a Voltage-Controlled Delay Block (VCDL), two 2x1 group selector, 4x2 group selector, and three NAND-type SR Latch.

A. WTA Block Diagram

In figure 7 the block diagram of this four-input voltage WTA is presented. This block requires four inputs as well as a CLK. CLKb is locally generated in the circuit with a simple inverter.

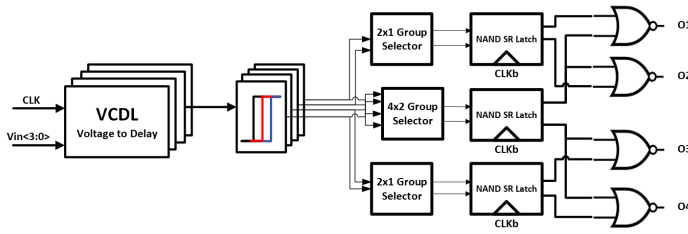


Fig. 7. Proposed WTA Block Diagram

B. Analysis on VCDL

The VCDL block is constructed in the following way (Figure 8). The input stage takes in the CLK signal as well as the V_{in} signal. The V_{in} signal turns into a current source that biases the branches of inverters in the following stage. An equivalent current is also copied in the PMOS stages by a diode-connected PMOS. This circuit's main goal is to turn CLK into a delayed clock that is proportional to the V_{in} values.

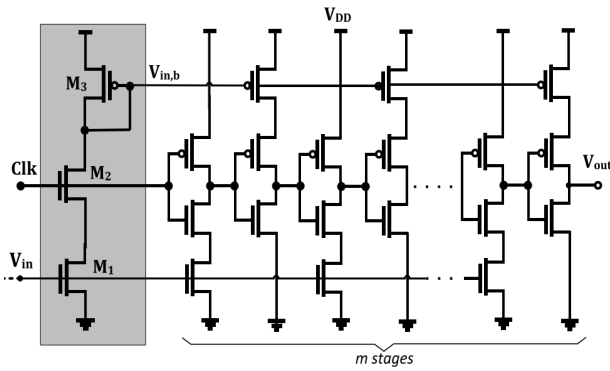


Fig. 8. VCDL Circuit Diagram

The function of mapping input voltage to delay is achieved by modulating the bias current through the inverters. In figure 9, we can see that for the N-stage current-starving inverter, the charge and discharge current is limited by the bottom NMOS, which acts as a current source that is positively correlated to input voltage. Since we are operating in subthreshold region,

the relationship is exponential, which has implications on speed and accuracy. This will be discussed later.

1) Delay Analysis

We can analyze the delay created by this single stage, which can give us insights on how stages of current-starving inverters have an effect on delay. First of all, we assume that CLK is alternates between VDD and 0. In the figure, I1 charges the capacitor (gate of next stage). Using the equation of subthreshold, we can get the following.

$$I_1 = i_{d,p} - i_{d,n}$$

$$= \frac{W}{L_p} \cdot K_p \cdot U_T^2 (n-1) e^{\frac{V_{DD}-V_{th}}{nU_T}} - \frac{W}{L_n} \cdot K_n \cdot U_T^2 (n-1) e^{\frac{V_{in}-V_{th}}{nU_T}}$$

If we assume this current remains constant, then we can estimate the delay using the basic equation of capacitor charging. A graph to illustrate this is also provided.

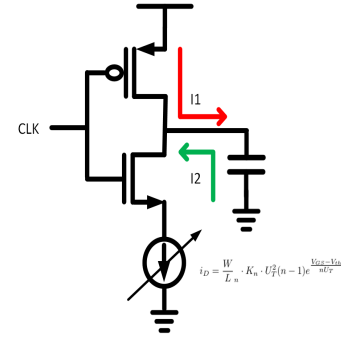


Fig. 9. VCDL Circuit Diagram

$$\Delta V_{DD} = \frac{\int_0^t i_D(v_{in}) dt}{C_L}$$

Therefore for an N-type current starving inverter, the rise and fall time is such.

$$t_{delay} = \Delta V_{DD} \cdot C_L \cdot \frac{1}{i_D(v_{in}) \cdot nU_T}$$

$$i_D(v_{in}) = \frac{W}{L} \cdot K_{n,p} \cdot U_T^2 (n-1) e^{\frac{v_{in}-v_{th}}{nU_T}}$$

From the non-linear nature of current in sub-threshold region, we can see how delay varies exponentially with input voltage. A relationship can be seen in figure 10.

In figure 10, we see that the further we go into sub-threshold region, the more time delay we see between signals that are 10mV apart from each other. However, the time required for the delay to arrive also increased exponentially, which can have

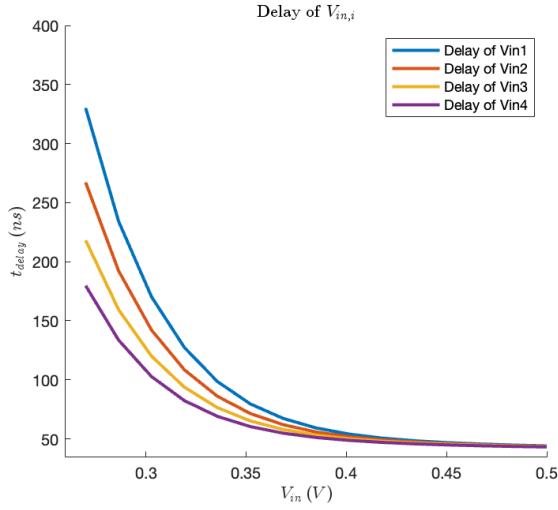


Fig. 10. VCDL Delay with Varying Voltage Input. $V_{in_{i+1}} = V_{in_i} + 10mV$ where $V_{in_1} = 270mV$

implications on our timing. For example, when the highest input voltage is at 270mV, then it will take more than 300ns for the delay clock to arrive, which means the maximum frequency that this circuit can run at is 3.3 MHz.

C. Analysis on 2x1 Selector

In this part, we will discuss the 2x1 group selector. The 2x1 group selector is a simple circuit which has PMOS inputs that are reset each cycle by CLKb signal. With the delayed clock edge from VCDL, the output is pulled up. In conjunction with the SR latch in the later stage, the circuit achieves a group selection function that selects the fastest output. Its function will be more evident in the case of 4x2 group selector. And with this concept, operations can be expanded for more inputs as well. A figure of the 2x1 group selector can be seen in figure 11.

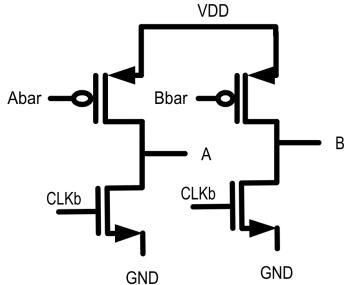


Fig. 11. 2x1 Group Selector

D. Analysis on 4x2 Group Selector

In this part, we will discuss the 4x2 group selector. The 4x2 group selector has 4 PMOS inputs that are reset each cycle by CLKb signal (See figure 13). With the delayed clock edge

from VCDL, the group that contains the first delay edge will be pulled up, which triggers the SR Latch to register. With the 2x1 group selector, we get to know the comparison between two signals. With the 4x2 group selector, we get to know the comparison between two groups of signals (in groups of 2). With these two information, we can deduce the fastest signal to find the winner. In figure 12, the group selection is demonstrated for better clarity.

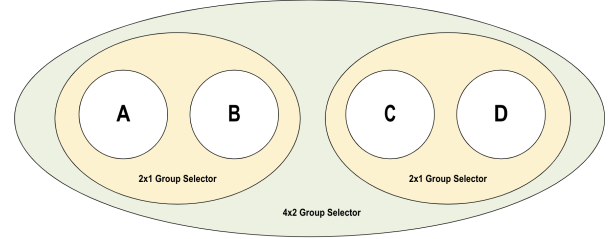


Fig. 12. Diagram of group selector function: 2x1 group selector selects the fastest signal of two inputs, 4x2 group selector selects the fastest signal in group of two without knowing which signal in the group was faster

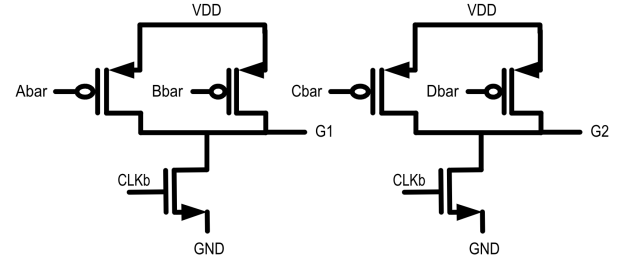


Fig. 13. 4x2 Group Selector

E. Analysis on SR Latch

This section discusses the SR Latch. In this work, the SR latch is implemented with NAND gates. A truth table can be seen below:

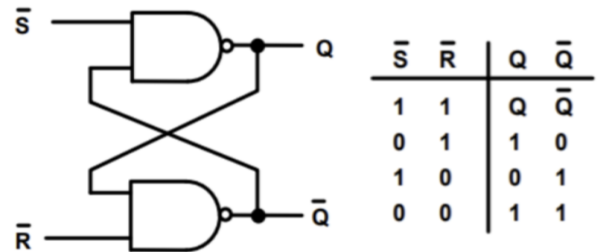


Fig. 14. NAND gate truth table

1) Delay Analysis

The important feature of this NAND phase detector is to turn the delay differences of the two inputs into binary values that reflects this order. i.e. $f_{NAND}(t_1, t_2) = [0, 1]$ for $t_1 < t_2$. This means that the output that corresponds to the first input edge turns to zero. However, when the difference δt between t_1 and t_2 are smaller, the time it takes for the NAND phase detector to register the correct value is longer. This is a common condition of meta-stability. In fig 15 we can see that when δt is 1ps, the time it takes for meta-stability to resolve is around 80ps, where as it is only 20ps in the case of a δt of 20ps.

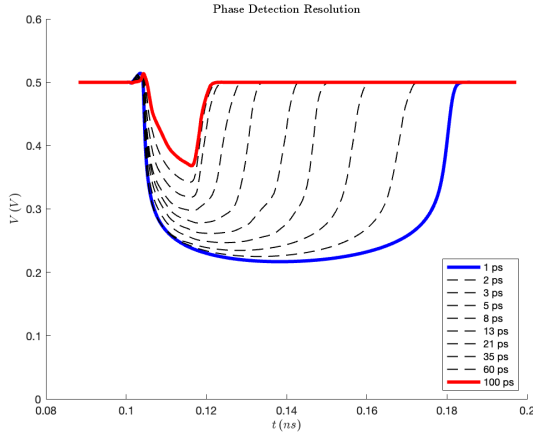


Fig. 15. TTTT

This phenomenon can be explained by the contention between the two input signals. It can be visualized in the block diagram in 16. Given the inputs (V_1 and V_2) we can see that they go through a positive feedback loop that turns output (O_1 and O_2) to their desired value (if V_1 arrives first then $O_1 = 0$ and $O_2 = 1$). This function is achieved by NMOS reducing the output voltage node which also triggers the PMOS connected on the other output to provide complementary signals. However, if the arrival time of V_2 is sufficiently close, then the current created by V_2 will counter the positive feedback, which slows down the time it take to resolve meta stability. Although theoretically, meta-stability will be resolved given sufficient time, it might violate timing constraint of the circuit.

2) Important Considerations

To improve the timing resolution of this phase detector, we can consider skewing the NMOS and PMOS in order to strengthen the positive feedback. One can either downsize the NMOS attached to the input signals which makes the feedback path relatively stronger.

F. Example Waveform of group selector and SR Latch

To provide more clarity on the group selector and its function with the latch, an example waveform is provided in figure 17. We can see that when an input arrives first (aka higher input

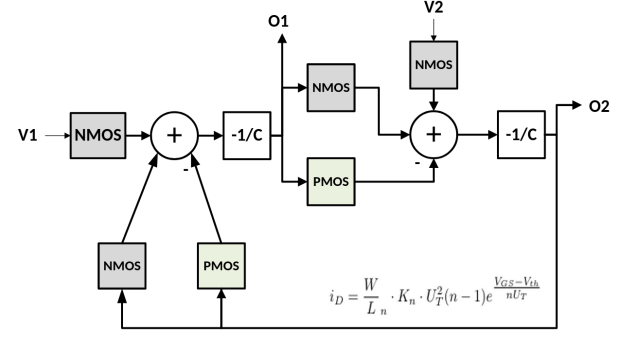


Fig. 16. Block Diagram of Cross-Coupled NAND

voltage), then the output of the latch of that corresponding input will drop to zero, while the other stays at VDD.

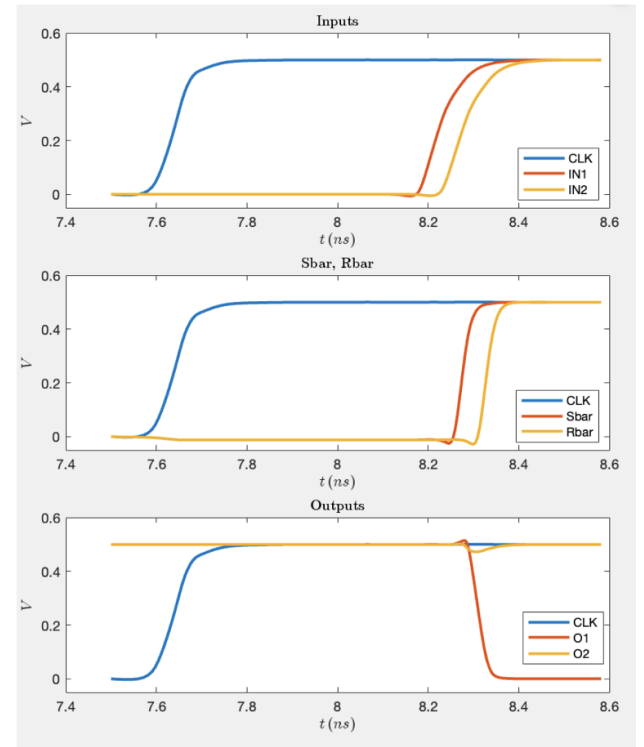


Fig. 17. An example output waveform of a 2x1 group selector with a SR NAND latch.

VI. SIMULATION RESULT

In this section, we will discuss the result that is simulated from the proposed circuit. Both the layout extracted values and the schematic level result will be provided and discussed. Monte-Carlo simulation is also run to ensure good stochastic performance.

A. Power Consumption

Power consumption is run under the condition of 100 clock cycles for the monte carlo simulation and 500 clock cycles for

single run simulation. Here are the result

TABLE I
POWER PERFORMANCE IN PRE-LAYOUT AND POST-LAYOUT

Metric	Pre-Layout		Post-Layout	
	Mean (μ)	STD (σ)	Mean (μ)	STD (σ)
Power (MC) (nW)	89.52	28.43	91.78	25.41
Power (Single) (nW)	127.5	-	131.5	-

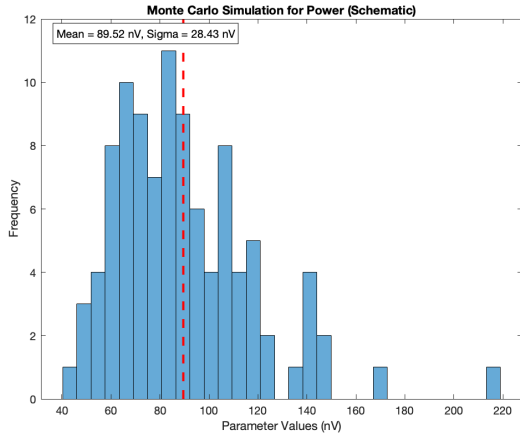


Fig. 18. Monte Carlo Simulation of Power on the Schematic Level

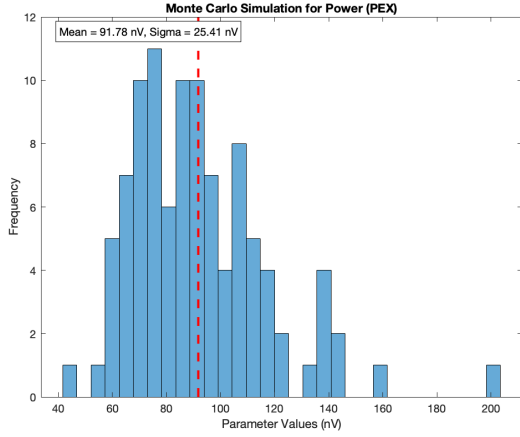


Fig. 19. Monte Carlo Simulation of Power on the Post-Layout Level

B. Delay

The delay is measured at when input voltage are sufficiently apart. We define it here at 5mV. Here are the result.

TABLE II
DELAY PERFORMANCE IN PRE-LAYOUT AND POST-LAYOUT

Metric	Pre-Layout		Post-Layout	
	Mean (μ)	STD (σ)	Mean (μ)	STD (σ)
Delay (Monte Carlo) (ns)	90	280	210	450

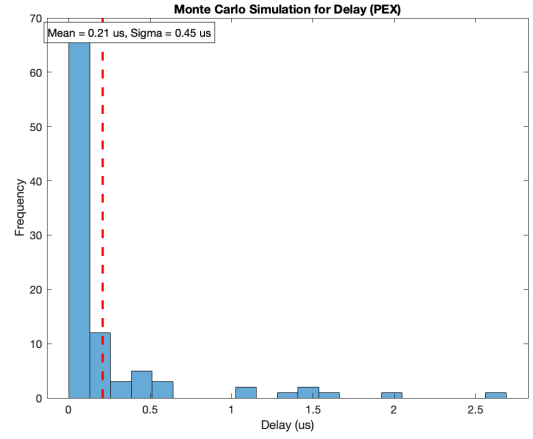


Fig. 20. Monte Carlo Simulation of Delay on the Post-Layout Level

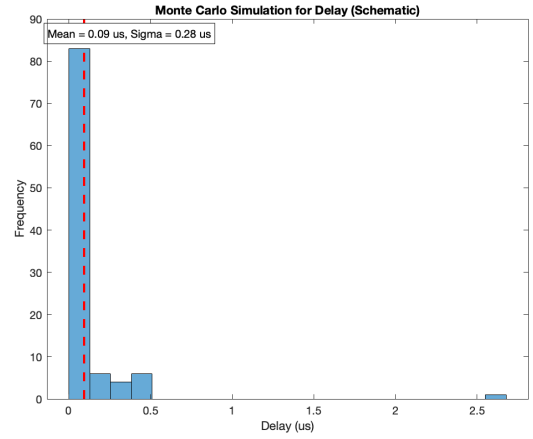


Fig. 21. Monte Carlo Simulation of Delay on the Schematic Level

C. Frequency

In this work, we aim to compare our result with [4], which is operating at 1MHz, thus, the clock frequency is also set at 1MHz.

D. Precision

Precision is measured using a ramp. We measure the minimum voltage that is required for the circuit to register the correct winner signal. The table below is the result.

TABLE III
PRECISION PERFORMANCE IN PRE-LAYOUT AND POST-LAYOUT

Metric	Pre-Layout		Post-Layout	
	Mean (μ)	STD (σ)	Mean (μ)	STD (σ)
Precision (MC) (mV)	1.11	1.57	1.27	1.54
Accuracy (MC) (%)	99.52	-	99.45	-

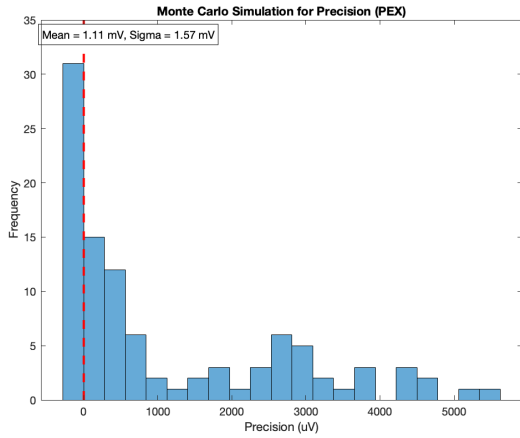


Fig. 22. Post-Layout Precision Level Result

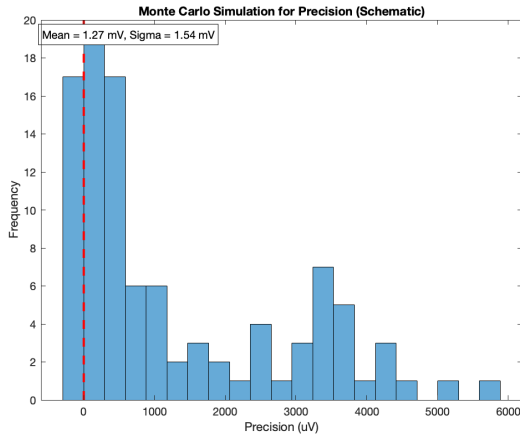


Fig. 23. Schematic Level Precision Level Result

E. Figure of Merit

Below are the equations for the figure of merit. A table of comparison between pre-layout and post-layout is also provided below.

$$FoM := \frac{Power}{f \cdot N}$$

$$FoM_{new} := \frac{Power}{f \cdot N} \cdot t_{delay} \cdot \gamma$$

TABLE IV
PRECISION PERFORMANCE IN PRE-LAYOUT AND POST-LAYOUT

Metric	Pre-Layout		Post-Layout	
	Mean (μ)	STD (σ)	Mean (μ)	STD (σ)
FoM ($\frac{\mu W}{MHz}$)	0.0318	-	0.0328	-
FoM_{new} ($FoM \cdot ns \cdot mV$)	3.176	-	8.748	-

F. Output Waveform

A picture of the output is shown in 24. The alternating colors in the output waveform is an overlay of four different signals. At each clock cycle, only one output signal will be at high while the rest stays at zero. We can observe that the output signal follows the winner of the input signal (subgraph 3), which means that the circuit achieves the correct functioning of a winner take all circuit.

G. Summary

A summary of all the important parameters of our WTA circuit is summarized below. A comparison of these parameters with other similar circuits will also be discussed later. We can see that most parameters remain similar in pre and post layout condition.

TABLE V
COMPARISON OF PRE-LAYOUT AND POST-LAYOUT

Metric	Pre-Layout		Post-Layout	
	Mean (μ)	STD (σ)	Mean (μ)	STD (σ)
Power (Monte Carlo) (nW)	89.52	28.43	91.78	25.41
Power (Single Run) (nW)	127.5	-	131.5	-
Precision (μV)	1.11	1.57	1.27	1.54
Accuracy (%)	99.52	-	99.45	-
Delay (Monte Carlo) (ns)	90	280	210	450
FoM	0.0318	-	0.0328	-
FoM_{new}	3.176	-	8.748	-

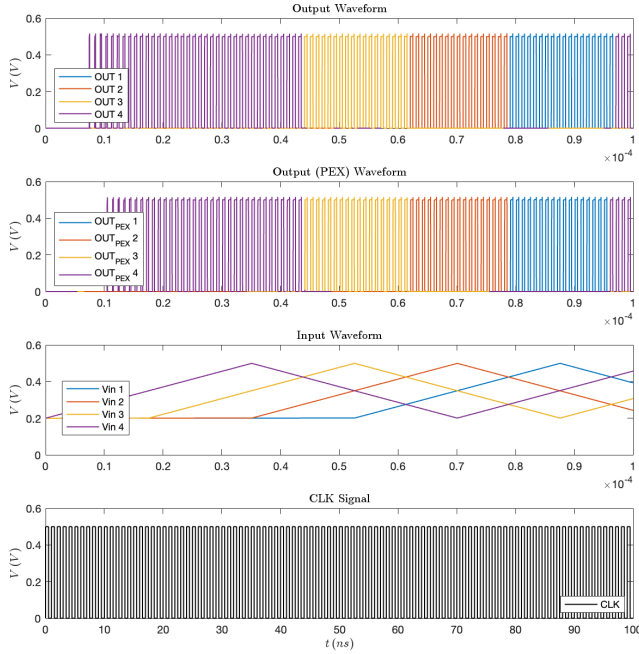


Fig. 24. Output waveform of proposed WTA circuit.

VII. CHIP LAYOUT DISCUSSION

The chip is layout in 180nm TSMC CMOS technology. All transistors are using nmos-mac model. Power distribution are deployed in metal 5 and metal 4 in grid like structure with double vias to Metal 1 for local supply.

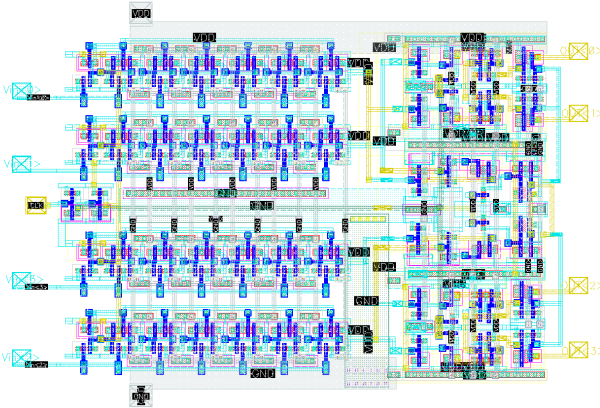


Fig. 25. Layout Demonstration of the Chip. The chip occupies an area of $29\mu\text{m} \times 41\mu\text{m}$

VIII. COMPARISON WITH RECENT WORKS

Below is a comparison between this work and a similar architecture. Since this time-domain voltage-mode WTA is a relatively new idea, only [4] is a reasonable comparison as it uses similar architecture.

TABLE VI
COMPARISON OF PERFORMANCES OF FOUR CIRCUITS

Metric	2019 [4]	This Work
Technology	130nm	180nm
Supply Voltage (μV)	0.5	0.5
Clock Frequenct (MHz)	1	1
Power Consumption (nW)	750	130
Precision ($m\text{V}$)	2.1	1.27
Accuracy (%)	99.3	99.45
Delay (ns)	-	210
Chip Area (μm^2)	840	1189
FoM	0.25	0.0328
FoM_{new}	-	8.748

IX. FURTHER IMPROVEMENT

Some further improvement of the proposed circuits are below:

- Improve amount of input numbers
- Insert Scan-Chain to reduce pads needed
- Explore voltage ranking/sorting mechanism
- Improve analog WTA error measurement method and figure of merit definition

A. Improve Input Numbers

Input numbers can be improved using the same logic in this circuit. An idea is to use two 8x4 group selector and four 2x1 group selector as well as a 3 input NOR gate. An example schematic of 8x4 group selector is in figure 26. With this 8x4 group selector, we can select the winner group of signal (in groups of 4). For example, given a group of signal [A,B,C,D,E,F,G,H], if the fastest group from ABCD v.s. EFGH is ABCD and the fastest group from ABGH v.s. CDEF is ABGH, then we can deduce that the fastest signal has to be either A or B. And with a 2x1 group selector, we can deduce the winner signal.

B. Scan-Chain Insertion

WTA circuit is considered a pad-limiting circuit where pad areas occupyes more space than the circuit itself, therefore, it is crucial to be able to reduce the amount pads needed. An idea that can be implemented is using a scan-chain. In this circuit, we can have the output be stored in 4 flip-flop in parrallel, which will be read out in series in the traditional "test" mode. With that, we can reduce the number of pads by 2. (4 ouputs - 1 outputs + 1 test mode signal)

C. Voltage Sorting

Voltage sorting was another direction of interest in the reach of WTA circuits. With our WTA, we are able to achieve a function as such:

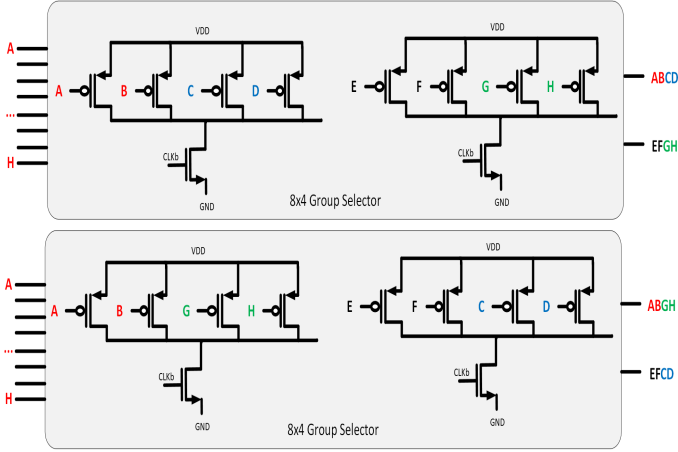


Fig. 26. Proposed Schematic for 8x4 group selector

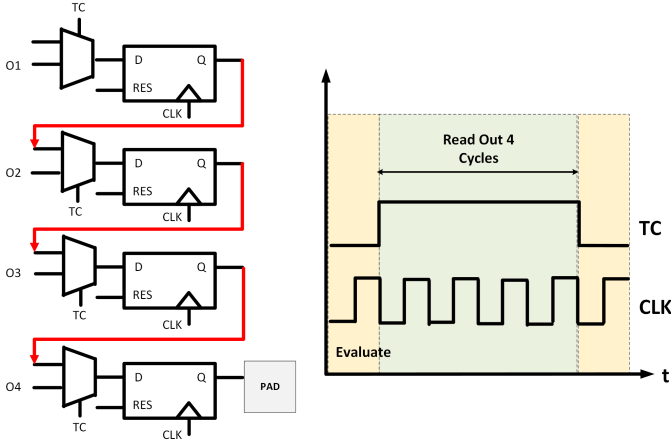


Fig. 27. Scan chain diagram for 4 input WTA test mode

$$f(x_1, x_2, x_3, x_4) = [0, 1, 0, 0]$$

Thus, we can use the following algorithm to achieve $O(n)$ complexity. We can simply use the output to remove the input that was deemed highest (x_2 in the example), and continue the comparison. We will stop when the amount of comparisons reaches $n-1$. The circuit can be implemented with another latch system that will disable selected input voltage until the end of each n -cycles when it is deemed as winner during the last cycle. A simple modification of the VCDL can be done to make it controllable. It is visualized in figure 28 and the waveform of a controlled VCDL is provided in figure 29.

D. Analog WTA Error Measurement

During this project, some time were dedicated as well to the research of novel analog WTA circuit methods. However, during the research, a gap in accurate definition of WTA error measurement is noticed. In order to allow reasonable comparison between future works, a definition of analog WTA precision

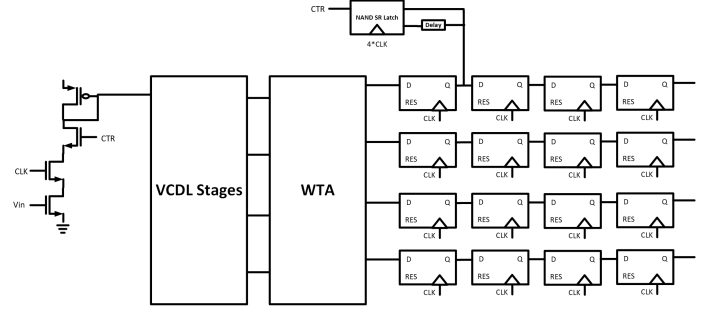


Fig. 28. Example of WTA voltage sorting as well as modified VCDL diagram

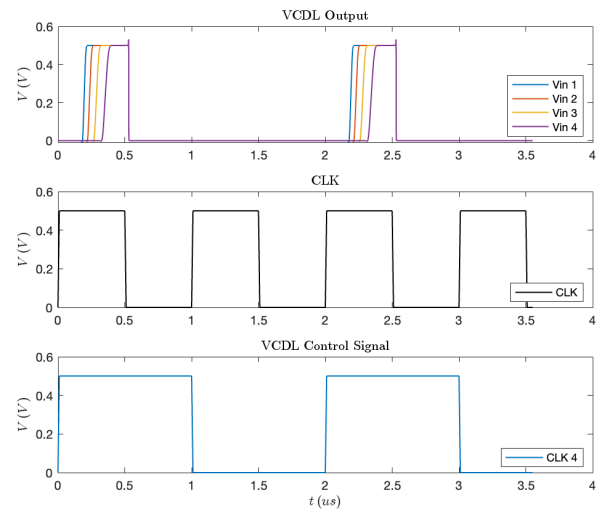


Fig. 29. Waveform of controlled-VCDL

must be developed. Since the parameter of analog circuits changes greatly across operating region (saturation, subthreshold saturation...) We might need to define multiple metrics for metrics like precision to better reflect the circuit's ability in different region.

X. CONCLUSION

In this paper, we proposed a new topology of a four-input time-domain WTA circuit that achieves high accuracy with more than 75% power reduction compare to previous similar work. This circuit leverages the digital-to-time architecture with positive feedback NAND latches to achieve fast response. Moreover, this circuit does not have an inherit asymmetry which provides higher accuracy than previous work. Silicon results will be confirmed to verify efficiency and efficacy after fabrication.

Future works will focus on generalizing this time-domain WTA architecture to expand number of inputs as well as improving allowable input voltage range while preserving power consumption level.

XI. APPENDIX: OTHER EXPLORATIONS

In the first part of this project, around three weeks of time were dedicated to exploring potential analog loser-take-all (LTA) architectures. Their schematics and mode of operations are provided below for completion.

A. Two Input Analog LTA

The idea of this circuit is related to the operation of transistors. A lower voltage signals will push the transistor below it to linear region, thus drastically reducing its V_{DS} value, using this V_{DS} and V_{GS} , we can recompute the A value.

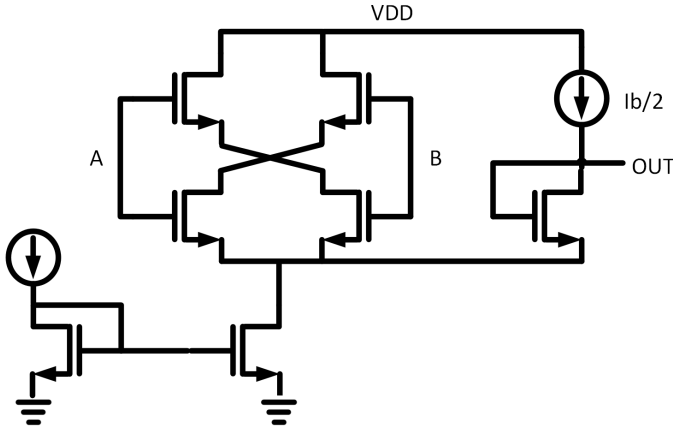


Fig. 30. Proposed schematic of a 2 input LTA

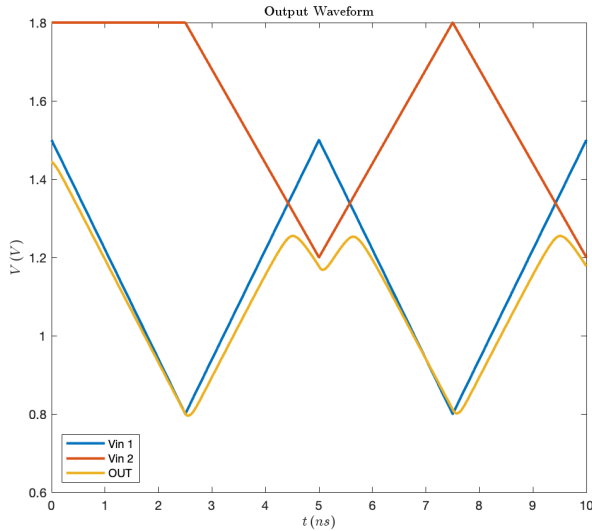


Fig. 31. Output waveform of the proposed 2 input LTA circuit

B. Three Input Analog LTA

We use similar inspirations from the two inputs LTA to further expand the input capability of this circuit. The circuit for 3

inputs LTA is proposed below in figure 32. As mentioned, the loser voltage will push the transistors below it to linear region, which drastically reduces their V_{DS} , making the output voltage highly correlated with the loser input voltage. On the other branches, since the transistors are in saturation, their V_{DS} is not highly related to their V_{GS} , thus decoupling the non-loser signal with the output signal.

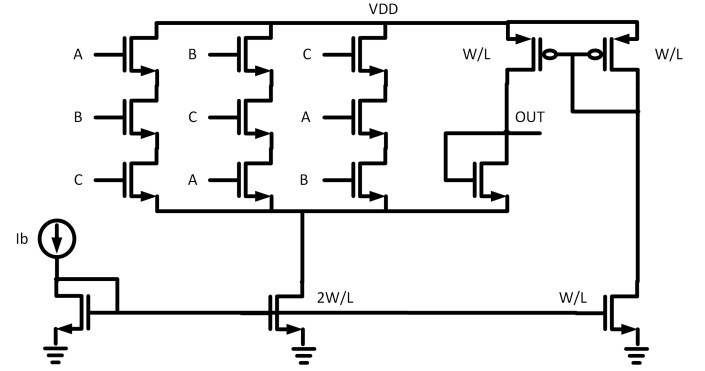


Fig. 32. Proposed schematic of a 3 input LTA

C. Four Input Analog LTA

For four input LTA, the circuit is proposed below. We have a DC path capacitor which serve as a blocking path of the winner signal. When the winner takes all the current, then the capacitor is charged, thus the output voltage is proportional to the loser voltage. A demonstration of its efficacy is proposed below in figure 34.

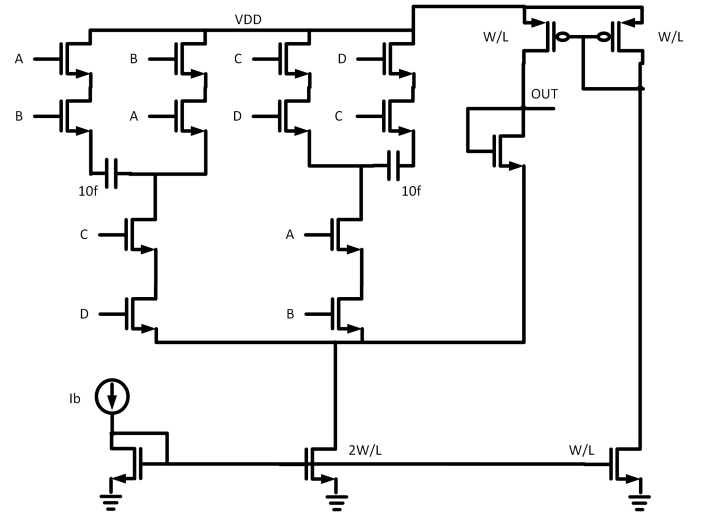


Fig. 33. Proposed schematic of a 4 input LTA

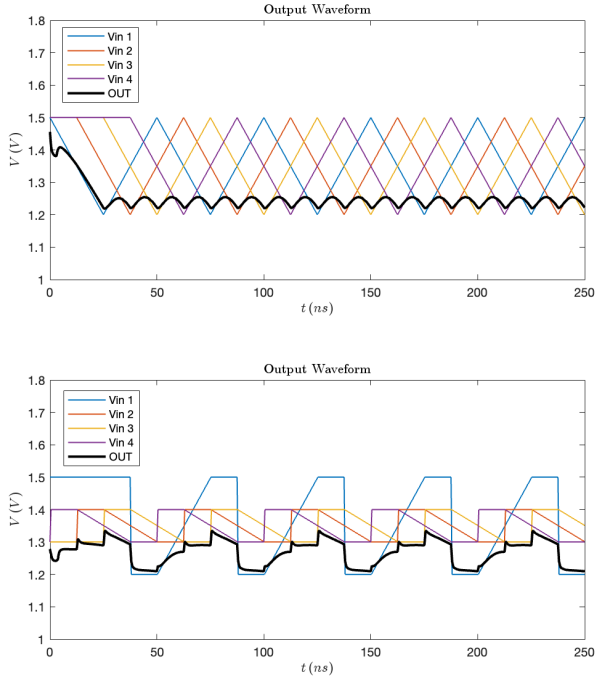


Fig. 34. Output waveform of the proposed 4 input LTA circuit

D. Summary

A summary of the proposed analog LTA circuits are provided below. As the emphasis of this work is digital domain, the performance of the analog circuits are not measured in detail. However, the result of the LTA circuits provide a potential pathway to a novel circuitry.

TABLE VII
INFORMATION OF PROPOSED ANALOG LTA CIRCUITS

Metric	2 Input	3 Input	4 Input
Bias Current (uA)	10	20	20
VDD	1.8	1.8	1.8
Input Range (V)	1.2-1.8	1.2-1.8	1.2-1.8

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